

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: B.Tech. **Branch :** Civil Engineering

Semester : V

Subject Code & Name: BTCVC502 Geotechnical Engineering

Max Marks: 60

Date: 08/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Dry soil is a ----- phase system a. one b. Two c. Three d. Four	1/1	1
2	The soil having particles of variety of sizes is said to be----- a. Well graded b. Gap graded c. Poorly graded d. Uniformly graded	1/1	1
3	The specific gravity of a soil solids can be determined by using-- a. Sieve b. Pycnometer c. Hydrometer d. Oven	1/1	1
4	The constant head permeability test is suitable for a. CG soil b. FG soil c. CG & FG soil d. None	1/2	1
5	The critical exit gradient for the seepage of water out of a soil is a. $\frac{G-1}{1+e}$ b. $\frac{G-1}{1-e}$ c. $\frac{G+1}{1+e}$ d. $\frac{e-1}{1+e}$	1/2	1
6	A flow net is drawn for a weir. The total head loss is 8m. Number of Potential drops are 8 and the length of flow of path of last flow field is 1m. The exit gradient is a. 0.5 b. 0.75 c. 1.0 d. 1.25	1/2	1
7	Deviatric pressure in triaxial cell is known as ----- pressure a. cell b. chamber c. Pore d. Additional vertical	1/3	1
8	A purely clayey soil is having unconfined compressive strength of 40kpa, its cohesion will be ----kPa a. 10 b. 20 c. 40 d. 80	1/3	1
9	Remoulded strength of soil having undisturbed strength 50ka and sensitivity of 2 is --- kPa	1/3	1

	a. 25	b. 50	c. 75	d. 100		
10	In a compaction test, as a compaction effort is increased, the optimum water content				1/4	1
	a. Decreases	b. Remains same	c. Increases	d. Increases first and then decreases		
11	In case of compaction, Dry density of soil is maximum when water content of soil is				1/4	1
	a. 0 %	b. 100 %	c. OMC	d. None		
12	Coefficient of compressibility is the slope of virgin portion of -----curve				1/4	1
	a. flow	b. compaction	c. $e - p$	d. $e - \log p$		
Q. 2	Solve the following.					12
A)	Derive relation $\gamma_d = \frac{G\gamma_w}{(1+e)}$ use 3 phase labelled diagram.				5/1	6
B)	Following observations were recorded during specific gravity test Find Specific gravity of soil. Mass of pycnometer = 40.1gm Mass of pycnometer and dry soil = 65.8gm Mass of pycnometer dry soil and water = 154.5gm Mass of pycnometer and water = 138.5gm				3/1	6
Q.3	Solve the following.					12
A)	With a neat labelled sketch explain plasticity chart. Explain meaning of each dual symbol used to classify the soil (Minimum six symbols)				4/2	6
B)	A falling head permeability tests was carried out on a 120cm long sample of silty clay. The diameter of sample and the stand pipe were 10cm and 1.0cm, respectively. the water level in stand pipe was observed to fall from 80 to 60cm in 12 min, determine the coefficient of permeability of the soil in m/day				4/2	6
Q.4	Solve Any Two of the following.					12
A)	Find the average coefficient of permeability soil mass made up of three horizontal layers. The first and second layers have the same thickness of 0.6m each. The third layer is 0.7m thick the coefficient of permeability of first, second, and third layers are 1×10^{-3} , 2.5×10^{-2} and 4.5×10^{-3} cm/s, respectively				4/2	6
B)	Direct shear test was conducted on a river sand using sample of size 60mm X 60mm X 25mm and following results were obtained determine shear parameters of soil					6

C)	Compare direct shear test and tri-axial shear test (minimum 6 points)	4/2	6
Q.5	Solve Any Two of the following.		12
A)	What is compaction of soil? Discuss with neat illustrative sketches factors affecting compaction of soil.	4/3	6
B)	Draw e – p curve for soil? Label different branches of this curve. What are the slopes of these branches?	4/4	6
C)	Calculate total active earth pressure and its position with respect to bottom of wall acting on a retaining wall of height 9m retaining two layered soil on back side of it. Top layer 4.2m thick $\gamma=19\text{kN/m}^3$, $c = 0$ and $\phi = 30^\circ$ followed by second layer having $\gamma=18\text{kN/m}^3$, $c = 0$ and $\phi = 27^\circ$	5/4	6
Q. 6	Solve Any Two of the following.		12
A)	The following data refer to a sample of soil: Percent passing 4.75 mm IS Sieve = 64 Percent passing 75- μ IS Sieve = 6 Uniformity Coefficient = 7.5 Coefficient of Curvature = 2.7 Plasticity index = 2.5 Classify the soil. Draw flowchart of classification to classify soil.	5/2	6
B)	Explain Rehmann method to calculate active earth pressure acting on wall with the help of illustrative sketch (use mathematical instruments to draw sketch – avoid freehand sketch)	5/4	6
C)	Draw typical grain size distribution curve for well graded soil. How do you calculate coefficient of curvature and coefficient of uniformity for the soil. What are the values of C_c and C_u for poorly graded soil?	5/1	6
	*** End ***		

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Supplementary Winter Examination – 2024

Branch : Civil Engineering

Semester : V

Subject Code & Name: CV503_Y17_Geotechnical Engineering

Course: T.Y. B.Tech

Max Marks: 60

Date: 11/02/2025

Duration: 3 Hr.

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		(Level/CO)	Marks				
Q. 1	Objective type questions. (Compulsory Question)		12				
1	The process of soil compaction results in:	1/CO1	1				
	<table border="1"> <tr> <td>a. Increase in volume</td> <td>b. Decrease in volume</td> <td>c. No change in volume</td> <td>d. Increase in permeability</td> </tr> </table>	a. Increase in volume	b. Decrease in volume	c. No change in volume	d. Increase in permeability		
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2	In Terzaghi's effective stress principle, the effective stress is equal to:	1/CO2	1				
	<table border="1"> <tr> <td>a. Total stress minus pore water pressure</td> <td>b. Total stress plus pore water pressure</td> <td>c. Pore water pressure only</td> <td>d. Zero</td> </tr> </table>	a. Total stress minus pore water pressure	b. Total stress plus pore water pressure	c. Pore water pressure only	d. Zero		
a. Total stress minus pore water pressure	b. Total stress plus pore water pressure	c. Pore water pressure only	d. Zero				
3	Which of the following is a type of shear test used in soil mechanics?	1/CO3	1				
	<table border="1"> <tr> <td>a. Unconfined compression test</td> <td>b. Triaxial compression test</td> <td>c. Vane shear test</td> <td>d. All of the above</td> </tr> </table>	a. Unconfined compression test	b. Triaxial compression test	c. Vane shear test	d. All of the above		
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4	Darcy's law is primarily concerned with:	1/CO2	1				
	<table border="1"> <tr> <td>a. Stress distribution in soil</td> <td>b. Permeability of soil</td> <td>c. Soil compaction</td> <td>d. Soil consistency</td> </tr> </table>	a. Stress distribution in soil	b. Permeability of soil	c. Soil compaction	d. Soil consistency		
a. Stress distribution in soil	b. Permeability of soil	c. Soil compaction	d. Soil consistency				
5	The void ratio is defined as:	1/CO1	1				
	<table border="1"> <tr> <td>a. Volume of water divided</td> <td>b. Volume of solids divided</td> <td>c. Volume of voids divided</td> <td>d. Volume of soil divided by</td> </tr> </table>	a. Volume of water divided	b. Volume of solids divided	c. Volume of voids divided	d. Volume of soil divided by		
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	by the volume of solids	by the volume of voids	by the volume of solids	the total volume		
6	The coefficient of consolidation is measured in:				1/CO2	1
	a. m ² /s	b. cm ² /s	c. cm ³ /s	d. m/s		
7	Soil classification is done based on:				1/CO1	1
	a. Particle size	b. Atterberg limits	c. Consistency limits	d. All of the above		
8	The degree of saturation in soil indicates:				1/CO1	1
	a. The volume of water present in the soil	b. The amount of air in the soil	c. The proportion of voids filled with water	d. The total volume of the soil		
9	The unconfined compression test is commonly used for:				1/CO3	1
	a. Cohesive soils	b. Cohesion less soils	c. Saturated soils	d. Partially saturated soils		
10	The consolidation of clayey soil is typically controlled by:				1/CO2	1
	a. Vertical stresses	b. Shear strength	c. Pore water pressure	d. Horizontal stresses		
11	In the Mohr-Coulomb failure criterion, the cohesion is represented by:				1/CO3	1
	a. The slope of the failure envelope	b. The intercept on the shear stress axis	c. The stress ratio	d. The angle of internal friction		
12	The standard proctor compaction test is used to determine:				1/CO1	1
	a. The maximum dry density of soil	b. The permeability of soil	c. The shear strength of soil	d. The void ratio of soil		
Q. 2	Solve the following.					12
A)	Define moisture content. Enlist various method of determination of moisture content and explain anyone method in detail.				2/CO1	6
B)	Explain the Following terms:				2/CO1	6

	1) Voids Ratio 2) Porosity 3) Degree of saturation 4) Water Content 5) Specific Gravity 6) Air Content		
Q.3	Solve the following.		12
A)	A soil sample has wet density 21 kN/m ³ and dry density of 19 kN/m ³ . If the specific gravity of soil is 2.60. Calculate the porosity, moisture content and degree of saturation. Assume unit weight of water = 10 kN/m ³	3/CO1	6
B)	What is mean by consistency of soil, Atterberg limit and explain in detail method to calculate the liquid limit of soil.	3CO1	6
Q. 4	Solve Any Two of the following.		12
A)	What are the different factors affecting permeability of soil	2/CO2	6
B)	A 1.20 m layer of the soil (G = 2.65 and porosity = 35%) is subject to an upward seepage head of 1.80 m. What depth of coarse sand would be required above the soil to provide a factor of safety of 2.0 against piping assuming that the coarse sand has the same porosity and specific gravity as the soil and that there is negligible head loss in the sand.	3/CO2	6
C)	Calculate the consolidation settlement for a clay layer given the following: thickness of the layer = 5 m, initial void ratio = 1.2, compression index = 0.3, and the applied stress is 100 kPa.	3/CO2	6
Q.5	Solve Any Two of the following.		12
A)	Explain the fundamental equation of shear strength of soil given by Coulomb.	2/CO3	6
B)	A cylindrical sample of soil having cohesion of 0.8Kg/cm ² and angle of internal friction of 20°, is subjected to a cell pressure of 1.0Kg/cm ² . Calculate maximum deviator stress at which the sample will fail and the angle made by failure plane with the axis of sample.	3/CO3	6
C)	Derive the equation for effective stress in a soil mass.	3/CO2	6
Q. 6	Solve Any Two of the following.		12

A)	State and explain the assumptions in 'Terzaghi's theory of one-dimensional consolidation'.	2/CO2	6
B)	Explain the difference between Standard Proctor test and Modified Proctor test.	2/CO1	6
C)	A gravity retaining wall retains 10 m of a backfill, $\gamma = 25 \text{ kN/m}^3$ $\phi = 20^\circ$ with a uniform horizontal surface. Determine the total active thrust per meter run of wall. Assume the wall interface to be vertical.	3/CO3	6
*** End ***			